Subject: Mathematics
Class: Class X (CBSE)

CBSE CLASS 10 – PERIODIC TEST 10
Chapter 1: Real Numbers

Time: 60 Minutes
MM: 30

Student Name: _____ **Roll No.:** _____ **Date:** _____

GENERAL INSTRUCTIONS:

1. All questions are compulsory. Internal choices are provided in question subparts.
2. **Section A** comprises 5 MCQs and 1 Assertion-Reason question of 1 mark each.
3. **Section B** comprises 2 Very Short Answer (VSA) questions of 2 marks each.
4. **Section C** comprises 2 Short Answer (SA) questions of 3 marks each.
5. **Section D** comprises 2 Long Answer (LA) questions of 5 marks each.
6. **Section E** comprises 1 Case Study based question of 4 marks with internal subparts.
7. Use of calculators is strictly prohibited. All mathematical steps must be clearly shown.

SECTION A: Multiple Choice & Assertion-Reason (6 × 1 = 6 Marks)

Q1. If $\text{LCM}(x, 18) = 36$ and $\text{HCF}(x, 18) = 2$, then the value of x is: [1]
(a) 2 (b) 3 (c) 4 (d) 6

Q2. If $\text{HCF}(a, 8) = 4$ and $\text{LCM}(a, 8) = 24$, then a is equal to: [1]
(a) 12 (b) 16 (c) 6 (d) 8

Q3. For what natural number n does 4^n end with the digit zero? [1]
(a) for all even n (b) for $n = 5$
(c) for no natural number n (d) for $n = 10$

Q4. If n is any natural number, then 12^n cannot end with the digit: [1]
(a) 2 (b) 4 (c) 6 (d) 0

Q5. If the LCM of two natural numbers is 3600, which of the following numbers CANNOT be their HCF? [1]
(a) 600 (b) 500 (c) 400 (d) 150

Q6. **Assertion (A):** If $\text{LCM}(a, b) = 182$ and $a \times b = 2366$, then $\text{HCF}(a, b) = 13$.
Reason (R): For any two positive integers a and b, $\text{HCF}(a, b) = \frac{a \times b}{\text{LCM}(a, b)}$. [1]
(A) Both (A) and (R) are true and (R) is the correct explanation of (A).
(B) Both (A) and (R) are true but (R) is not the correct explanation of (A).
(C) (A) is true but (R) is false. (D) (A) is false but (R) is true.

SECTION B: Very Short Answer Type (2 × 2 = 4 Marks)

Q7. Prove that $\frac{1}{\sqrt{2}}$ is irrational, given that $\sqrt{2}$ is irrational. [2]

Q8. Prove that $7\sqrt{5}$ is an irrational number, given that $\sqrt{5}$ is irrational. [2]

SECTION C: Short Answer Type (2 × 3 = 6 Marks)

Q9. Prove that $\sqrt{p}$ is irrational for any prime p. [3]

Q10. If the HCF of 210 and 55 is expressible in the form $210 \times 5 + 55y$, find the value of y. [3]

SECTION D: Long Answer Type (2 × 5 = 10 Marks)

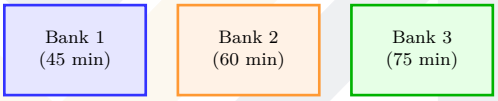
Q11. Three athletes A, B, and C start running around a circular track of circumference 1200 m at speeds of 6 m/s, 4 m/s, and 3 m/s respectively from the same starting point and in the same direction. (i) Find the time taken by each athlete to complete one full lap. (ii) After how many seconds will all three athletes meet again at the starting point? (iii) How many laps will each athlete have completed by that time? [5]

Q12. In an industrial plant, three automated sirens sound at intervals of 15 minutes, 20 minutes, and 25 minutes respectively. A fourth master siren sounds every 30 minutes. If all four sirens sound simultaneously at 6:00 AM: (i) Find the exact time when they will next sound simultaneously. (ii) How many times will all four sound simultaneously between 6:00 AM and 6:00 PM (including both endpoints)? (iii) Which siren sounds the greatest number of times during this 12-hour period, and how many times? [5]

SECTION E: Case Study Based Question (1 × 4 = 4 Marks)

Q13. CASE STUDY: CASE STUDY 10: OFF-GRID SOLAR ENERGY STORAGE BATTERY CHARGE CYCLES [4 Marks]

An off-grid solar-powered eco-resort in Ladakh uses three banks of lithium-iron-phosphate (LiFePO4) battery arrays to store solar photovoltaic energy. Due to varying capacities and charging regulators, Bank 1 completes a full charge-discharge cycle in 45 minutes, Bank 2 in 60 minutes, and Bank 3 in 75 minutes. The plant supervisor synchronizes all three battery banks to start charging at 7:00 AM.



Synchronization Interval = **LCM**$(45, 60, 75) = 900$ **min** $= 15$ **h**

(i) Find the least common multiple (LCM) of 45, 60, and 75 minutes. [2]

(ii) After how many hours from 7:00 AM will all three battery banks complete their cycles simultaneously again? [1]

(iii) Express 75 as a product of its prime factors in exponential form. [1]

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